PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2025-2026 | FIFTH SEMESTER B.TECH

SYSTEM VALIDATION REPORT

(System, Non-Functional and Security Testing)

PROJECT: SPRINTFLOW – PROJECT MANAGEMENT SYSTEM

(Workflow, Visibility, Estimation, and Kanban)

SUBMITTED BY: TEAM 11 (SECTION 5C)

Student Name	USN	Testing Role (STP Section 9)	Report Owner For
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Version: 1.0 (Final Submission) | Date: 06-11-2026 | Testing Phase: 02-11-2026 to 06-11-2026

Repository: https://github.com/Darshanpawar7/SE-SRS

Document: System Validation Report, Testing Phase (Part 3, item 2.b.ii)

Report owner: Dhatri Shivaprasad, Test Engineer (Frontend) (GitHub issue #92)

Prepared by: Team 11 (Section 5C, PES University)

Date: 06-11-2026

Build under test: `main` branch, commit `f9e0727`; defect retest on PR #98, commit `e8f5905`; final regression on `main` at commit `2b8fdbc`

Repository location: `docs/test-reports/SYSTEM_VALIDATION_REPORT.md`, test script `tests/system/system-validation.mjs`, raw results in `docs/test-reports/evidence/`

Revision History

Version	Date	Author	Change Summary
1.0	06-11-2026	Team 11	System test results for every case in the Test Case Specification, the STP security checks, cross-browser results, one defect with its retest, and the requirements traceability

1. Introduction

1.1 Purpose

This report checks the complete SprintFlow system against its requirements. Every case in the Test Case Specification was run end to end in real browsers against the production build, along with the security checks listed in STP Section 5.4. It covers the system test level of the Software Test Plan (Section 5.1) and the non-functional cases in STP Section 13.2.

1.2 Scope

In scope are the 26 cases in the Test Case Specification (20 functional and 6 non-functional), two supplementary cases we added for SRS requirements that had no case, and five security checks. Unit and integration results are in the Unit Test Report; code and branch coverage are in the Code Coverage and Branch Coverage Report.

1.3 References

- Software Test Plan, STP v1.0 (`docs/STP_Document_Team11.md`), Sections 5, 11, 13 and 14
- Test Case Specification (`docs/TEST_CASE_SPECIFICATION.md`)
- Software Requirements Specification (`docs/SRS_Document_Team11.md`)
- Sprint 1 deviations and ID mapping (`docs/SPRINT1_DEVIATIONS.md`)
- Earlier defects DEF-01 to DEF-03 (`docs/TEST_EXECUTION_AND_CICD.md`)

2. Test Environment and Approach

Item	Value
Build under test	`main` at commit `f9e0727`, built with `npm run build` and served with `vite preview`
Data mode	Zero-configuration (LocalStorage) mode with the seed data in `src/services/mockData.js`. No Supabase credentials were supplied, so no test wrote to a real database.
Browsers	Google Chrome 157.0.8103.38 (primary), Microsoft Edge 156.0.4314.58, Firefox 142.0.1 (Playwright build)
Screen size	1440 x 900

Item	Value
Test driver	Playwright 1.56.1, script `tests/system/system-validation.mjs` (`npm run test:system`)
Main run	06-11-2026, 17:37 IST to 17:39 IST, one uninterrupted run
Defect retest	06-11-2026, 21:37 IST, build of PR #98 (commit `e8f5905`), Chrome
Final regression	06-11-2026, 23:05 IST, final `main` (commit `2b8fdbc`), Chrome, Edge and Firefox
Raw results	`docs/test-reports/evidence/system-validation-results.json`

The specification marks most cases as manual. For this phase we wrote each manual step as a browser action in the Playwright script, and each expected result as a check that compares it with what the page actually shows. This made the run repeatable, and the actual result of every expected result is recorded rather than ticked by hand. The rules we followed:

- Every case starts in a fresh browser context: empty storage, the seed data, signed in as the Team Lead (Darshan P Pawar). The order of the cases cannot change the results.
- A case passes only if every one of its checks passes, as Section 2 of the specification requires.
- A case whose precondition cannot be met is recorded as Blocked, together with any steps that could still be run.
- A screenshot is saved for every failure.

3. Execution Summary

Group	Cases	Pass	Fail	Blocked
Test Case Specification, functional (TC-AUTH-01 to TC-LOG-01)	20	17	1	2
Test Case Specification, non-functional (TC-PERF-01 to TC-RECOV-01)	6	5	0	1
Supplementary cases for SRS requirements with no specified case	2	2	0	0
Security validation, STP Section 5.4	5	4	0	1
Total	**33**	**28**	**1**	**4**

Metric (STP Section 14.1)	Result	Target
Cases executed in full (pass + fail)	29 of 33 (87.9%)	100%
Cases with at least one step executed	31 of 33 (93.9%)	–
Pass rate of executed cases, main run	28 of 29 (96.6%)	95% or higher
Pass rate after the DEF-04 fix was retested	29 of 29 (100%)	95% or higher
Pass rate in the final regression on `main` (Section 12)	29 of 29 (100%)	95% or higher
Open critical or high-severity defects	0	0

Verdict: Pass for the zero-configuration system. Both defects found were fixed and retested, and four cases wait on a Supabase test project. Every requirement that can be exercised without a cloud database passed in the main run, apart from one usability detail in filtering (DEF-04, Section 5). Dhatri fixed DEF-04 in PR #98, and a wording slip found in the workload panel (DEF-05) was fixed in PR #101; both retests passed. A full regression run on the final build, in all three browsers, passed with no failures (Section 12). The cases that need a live Supabase project are listed in Section 6 and tracked in issue #96.

4. Results by Test Case

Requirement IDs follow the SRS, our baseline. The Test Plan ID for the same feature is shown next to it (mapping in `docs/SPRINT1_DEVIATIONS.md`, Section 1). The full expected-versus-actual result of every check is in Appendix A.

#	Test case	SRS ID	Test Plan ID	Title	Result	Key evidence (actual result)
1	TC-AUTH-01	PMS-F-001	PMS-F-001	Authenticate and establish session	**Pass**	Top bar shows Darshan / Team Lead; still signed in after refresh
2	TC-PROJ-01	PMS-F-002	–	Show project metadata (supplementary)	**Pass**	Key PMS, project name and active sprint in the top bar; project dates 2026-09-01 → 2026-10-30
3	TC-TASK-01	PMS-F-006	PMS-F-002	Create a task with complete metadata	**Pass**	Card shows title, High badge, 5 SP and Dhatri; with Backlog chosen it tops Backlog; survives refresh (default column: Section 7)
4	TC-TASK-02	PMS-F-006	PMS-F-016	Assign Critical priority	**Pass**	Badge reads "Critical" as text
5	TC-TASK-03	PMS-F-006	PMS-F-002	Reject a task with an empty title	**Pass**	Empty: browser validation: "Please fill out this field."; one space: inline message: "Enter a task title and choose a valid priority."; 6 cards before and after
6	TC-LABEL-01	PMS-F-007	–	Add labels to a task (supplementary)	**Pass**	Pills #feature #ui #api; unchanged after refresh
7	TC-KAN-01	PMS-F-003	PMS-F-003	Render the five-column board	**Pass**	Backlog, To Do, In Progress, Code Review, Done; every count badge matches its cards
8	TC-KAN-02	PMS-F-004	PMS-F-004	Transition a task by drag-and-drop	**Pass**	Card in In Progress; Backlog 1 → 0, In Progress 2 → 3; kept after refresh
9	TC-KAN-03	PMS-F-005	PMS-F-005	Advance a task by single action	**Pass**	To Do → In Progress; 0 enabled advance controls in Done
10	TC-KAN-04	PMS-F-004	PMS-F-004	Cancel a drag outside any column	**Pass**	1 dragstart, 0 drop events; card stayed in In Progress
11	TC-FILT-01	PMS-F-017	PMS-F-011	Filter by priority	**Fail**	An explicit empty state names the active filters: five columns each show "Drop tasks here"; no message names the active filters
12	TC-EST-01	PMS-F-008	PMS-F-006	Assign Fibonacci story points	**Pass**	Choices 1, 2, 3, 5, 8, 13, 21; each value saved; capacity 33 → 49 (task 5 → 21)
13	TC-EST-02	PMS-F-009	PMS-F-008	Calculate sprint capacity	**Pass**	Capacity 16 with 3+5+8, then 11 after removing the 5
14	TC-SPR-01	PMS-F-010	PMS-F-007	Create a sprint	**Pass**	Sprint 3 listed with goal and range 2026-11-06 → 2026-11-19; offered in the task dialog
15	TC-BURN-01	PMS-F-011	PMS-F-009	Update the burndown chart	**Pass**	Today's actual 20 → 15 (task 5 SP); ideal line unchanged
16	TC-DASH-01	PMS-F-012	PMS-F-010	Display KPI metrics	**Pass**	total 33, delivered 13, completion 39%, remaining 20; matches the board
17	TC-WORK-01	PMS-F-013	PMS-F-017	Show workload distribution	**Pass**	10 + 8 + 5 + 10 = 33; sprint total 33
18	TC-DOCS-01	not in SRS	PMS-F-018	Render in-app documentation	**Pass**	All three tabs render (3172 characters, 1093 characters, 947 characters)
19	TC-SUPA-01	PMS-F-015	PMS-F-013	Persist to Supabase PostgreSQL	**Blocked**	Precondition not met: needs a Supabase test project with database/schema.sql (and the 20261103 migration) applied.
20	TC-OFF-01	PMS-F-016	PMS-F-014	Persist offline without configuration	**Pass**	2 of 2 present after reopening the tab; top bar shows Local / Demo Mode
21	TC-SET-01	PMS-F-015	PMS-F-015	Switch data mode from settings	**Blocked**	Step 3 passed: malformed URL rejected, offline mode kept. Steps 1–2 need a Supabase test project (Section 6)
22	TC-LOG-01	PMS-F-014	PMS-F-012	Record an audit trail for mutations	**Pass**	Records 4 → 7: CREATE, UPDATE, DELETE, each with user, task and timestamp

#	Test case	SRS ID	Test Plan ID	Title	Result	Key evidence (actual result)
23	TC-PERF-01	PMS-NF-001	PMS-NF-001	Drag-and-drop transition under 50 ms	**Pass**	10 measured drops after 2 warm-ups: p50 2.3 ms, p95 6.6 ms, max 6.6 ms
24	TC-AVAIL-01	PMS-NF-002	PMS-NF-002	Stay usable with the network disabled	**Pass**	With the network off: every view opens, a task is created, top bar shows Local / Demo Mode
25	TC-COMPAT-01	not in SRS	PMS-NF-003	Core workflows on Chrome, Edge and Firefox	**Pass**	Same results and 0 px layout drift; chrome 157.0.8103.38, edge 156.0.4314.58, firefox 142.0.1
26	TC-SEC-01	PMS-SR-003	PMS-NF-004	Encrypted transport to Supabase	**Blocked**	An http:// project URL is refused. Live HTTPS traffic not inspected: needs Supabase (Section 6)
27	TC-LOAD-01	not in SRS	PMS-NF-005	Board with 100+ tasks	**Pass**	Board ready 162 ms after navigation with 106 tasks; drop-to-render p95 8.9 ms over 5 drops
28	TC-RECOV-01	PMS-NF-004	PMS-NF-006	No data loss across disconnect and reconnect	**Pass**	Tasks created offline still present after reconnect and reload (2 of 2 present)
29	SEC-XSS	PMS-SR-004	STP 5.4	XSS input sanitization	**Pass**	Script tags and event handlers in title, description and sprint goal never ran; rendered as text
30	SEC-RLS	PMS-SR-001	STP 5.4	Row Level Security on direct API calls	**Blocked**	Needs a live Supabase project to send cross-project API calls. The policies in database/schema.sql are checked statically by src/tests/databaseSecurity.test.js.
31	SEC-TOKEN	not in SRS	STP 5.4	Session token handling	**Pass**	Neither the password nor any JWT is kept in browser storage
32	SEC-KEYS	PMS-SR-002	STP 5.4	API key exposure in the client bundle	**Pass**	No complete JWT or Supabase secret key in the bundle (0 found in 2 files)
33	SEC-FUZZ	PMS-SR-004	STP 5.4	Input validation fuzzing	**Pass**	5,000-character title bounded (120 characters stored), no crash; special characters shown as typed

5. Defects Found

Defect IDs continue the series in `docs/TEST_EXECUTION_AND_CICD.md` (DEF-01 to DEF-03).

ID	Found by	Severity	Summary	Issue / fix	Status
DEF-04	TC-FILT-01, step 2	Low (usability)	When the board's filters match no task, the five columns show "Drop tasks here" and nothing says that filters are hiding the work, or which filters are active. "Reset Filters" is shown and restores the board.	Issue #95; fixed by Dhatri Shivaprasad in PR #98	Fixed and retested; PR #98 merged on 06-11-2026 and issue #95 closed
DEF-05	Review of TC-WORK-01 evidence	Cosmetic	The Capacity & Workload panel reads "1 tasks" for a member with one task, because `WorkloadList.jsx` always prints the plural. TC-WORK-01 still passes; the totals are correct.	Issue #100; fixed in PR #101	Fixed and retested (Section 12.1); issue #100 closed

5.1 DEF-04 in detail

Expected (TC-FILT-01): "an explicit empty state names the active filters and offers to clear them; columns are not left silently blank."

Actual (main run): the High priority filter itself works; step 1 passed with 3 High tasks shown. After a search term that matches nothing was added in step 2, the board looked exactly like an empty sprint (Figure 1). The Work items view already shows "No work items match." in the same situation, so the board was the only place affected. No data was lost or changed.

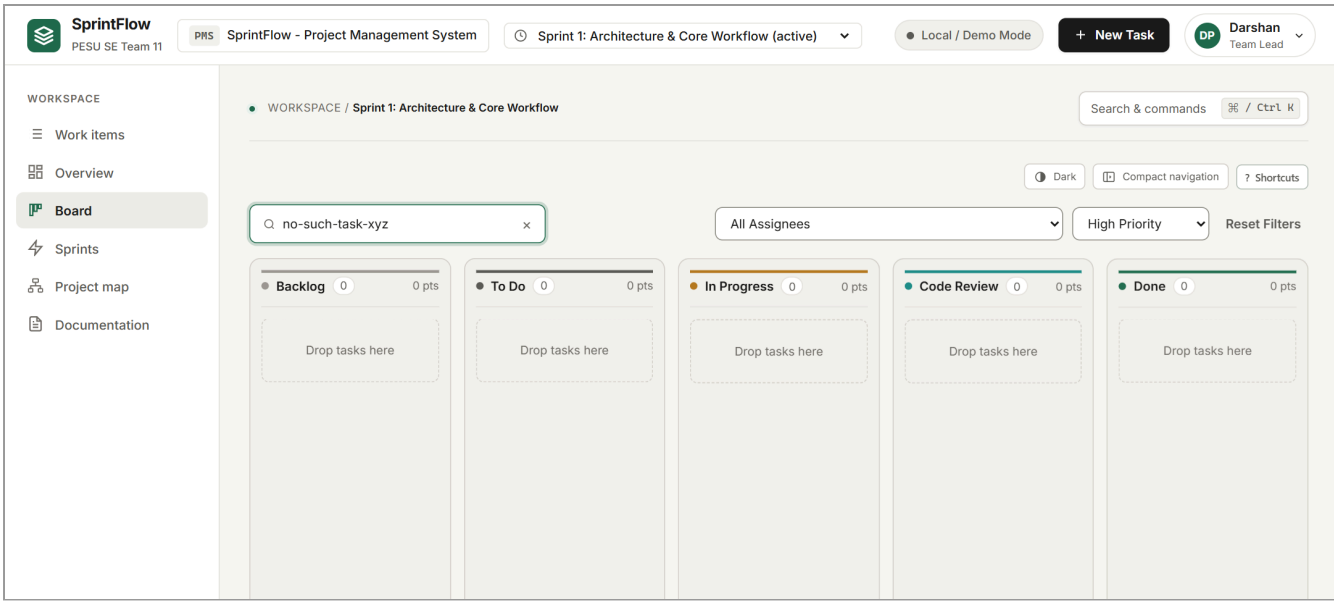


Figure 1: DEF-04 on the tested build (f9e0727): no message, five empty columns.

5.2 Retest of the fix (PR #98)

PR #98 adds a status message above the board that names the active filters and offers a Reset Filters button. It also adds three unit tests and extends the existing reset test. We built the PR (commit `e8f5905`) and ran TC-FILT-01 again in Chrome on 06-11-2026. All four checks passed:

Expected result checked	Actual result	Verdict
Only High priority tasks are visible	3 visible: high, high, high	**Pass**
No task matches the combined filters	0 cards visible	**Pass**
An explicit empty state names the active filters	Message shown: "No tasks match the active filters: Priority: High; Search: “no-such-task-xyz”." (Figure 2)	**Pass**
The empty state offers to clear the filters	"Reset Filters" button shown	**Pass**

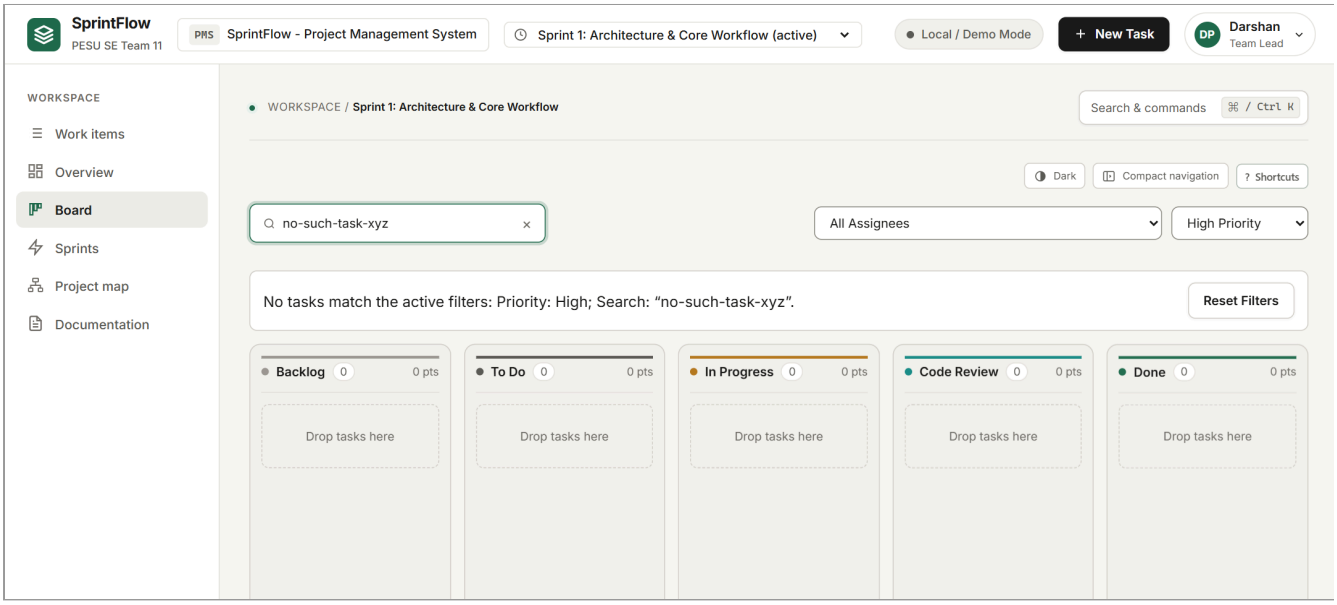


Figure 2: Retest on the fix build (PR #98): the empty state names the filters and offers Reset Filters.

To make sure the fix broke nothing else, we ran the whole Chrome suite on the same build: 29 passed, 0 failed and 4 blocked, the same four cloud cases as in the main run. On the fix branch, `npm test` passed 237 of 237 tests (the 234

existing tests plus the new ones), and lint and the production build both succeeded.

5.3 DEF-05

While checking the screenshots for this report we noticed that the workload panel says "1 tasks • 8 story points assigned" for Dhatri and "1 tasks • 5 story points assigned" for Chandan (workload screenshot in Appendix B). The text is built as `{taskCount} tasks` on lines 20 and 41 of `WorkloadList.jsx`. It does not affect any number, so TC-WORK-01 passes, but it is visible to every user. It was logged as issue #100 and fixed in PR #101 (retest in Section 12.1).

No other defects were found. No uncaught JavaScript error occurred in any case, the XSS and fuzzing inputs neither ran nor crashed the page, and every change survived a page refresh.

6. Blocked Cases: Supabase Test Project Needed

These cases have a live Supabase project as their precondition (the STP lists this as a dependency in Section 11). We did not run them against the team's live database, because their steps write rows to it. The table shows what could be checked without one; the rest is tracked in issue #96 and will be run on a separate Supabase test project.

Case	SRS	Verified now	Still to run on a Supabase test project	Supporting automated evidence
TC-SUPA-01	PMS-F-015	–	Create a task while connected, confirm the row in `tasks`, refresh and confirm it remains	`supabaseSync.test.js` (15 tests against a mocked client: reads, first-connection publish, column mapping, deletes, audit rows, failure handling)
TC-SET-01	PMS-F-015	Step 3: a malformed URL is rejected with "Enter a valid https:// Supabase project URL…", and the offline configuration is kept	Steps 1–2: valid credentials switch the indicator to cloud mode	`configSecurity.test.js`, `supabaseSync.test.js` (TC-SET-01 malformed-URL test)
TC-SEC-01	PMS-SR-003	A plain `http://` project URL is refused, so the client can only be configured for HTTPS	Inspect live requests: every request is HTTPS and no key appears in a URL	`configSecurity.test.js` ("accepts only HTTPS URLs…")
SEC-RLS	PMS-SR-001	–	From a second account, query another project's tasks through the REST API and expect zero rows	`databaseSecurity.test.js` ("enables RLS on every application table…")

Before these are run, `database/schema.sql` and then `database/migrations/20261103_work_signals.sql` must be applied to the test project. The migration has been needed since PR #87; without it every cloud task save is rejected.

7. Differences Between the Specification and the Built System

These are not defects. Each one is a documented design change or a detail the specification leaves open. They will be carried into the update of the Test Case Specification (Part 3, item 1, "Test Cases Updation").

#	Case	Specification says	System does	How it was tested
1	TC-TASK-01	New card appears at the top of Backlog	A new task starts in **To Do** unless another status is chosen (documented in `SPRINT1_DEVIATIONS.md`)	Created with the default (card lands in To Do) and again with Backlog chosen (card tops Backlog). All other expected results checked on both.
2	TC-AUTH-01	Select the Team Lead profile and confirm	Sign-in is a profile plus its password (documented in `SPRINT1_DEVIATIONS.md`)	Signed in as Darshan P Pawar (Team Lead) with the demo password
3	TC-TASK-03	Both attempts rejected with an inline validation message	Empty title: the browser's required-field message ("Please fill out this field."). Single space: the app's inline message.	Both rejected and no task created, so the case passes. The wording should name both mechanisms.

#	Case	Specification says	System does	How it was tested
4	TC-BURN-01	"Note the actual burn line"	The actual line is recorded once a day per browser, so a fresh browser has a single point	Read today's value from the chart's "Daily burndown values" table, before and after completing a 5-point task
5	TC-EST-02	"Remove the 5-point task"	–	Removed by deleting the task
6	TC-PERF-01	Measure with the Performance API	–	`performance.mark` at the drop event, measured to the DOM update that shows the card in its new column; 10 samples after 2 warm-ups
7	TC-AVAIL-01, TC-RECOV-01	Disconnect the network	–	Network disabled in the browser context, in zero-configuration mode. The cloud retry queue across a reconnect is covered by `persistenceRecovery` and `recovery` unit tests.
8	TC-COMPAT-01	Chrome, Firefox and Edge	–	Desktop Firefox is not installed on the test machine, so Playwright's Firefox 142 build was used
9	Section 10 of the specification	Lists 2 automated test files	35 unit test files (234 tests) plus this system suite	–
10	Section 11 of the specification	–	SRS PMS-F-002 (project metadata) and PMS-F-007 (labels) have no system case	Supplementary cases TC-PROJ-01 and TC-LABEL-01 were added; both pass

8. Requirements Traceability (SRS Baseline)

SRS ID	Requirement	Verified by	Result
PMS-F-001	Sign-in and profile switching	TC-AUTH-01	Pass
PMS-F-002	Project metadata	TC-PROJ-01 (supplementary)	Pass
PMS-F-003	Five Kanban columns	TC-KAN-01	Pass
PMS-F-004	Drag and drop	TC-KAN-02, TC-KAN-04	Pass
PMS-F-005	One-click stage advance	TC-KAN-03	Pass
PMS-F-006	Create and update tasks	TC-TASK-01, TC-TASK-02, TC-TASK-03	Pass
PMS-F-007	Labels and tags	TC-LABEL-01 (supplementary)	Pass
PMS-F-008	Fibonacci story points	TC-EST-01	Pass
PMS-F-009	Planned points and completion	TC-EST-02, TC-DASH-01	Pass
PMS-F-010	Sprint lifecycle	TC-SPR-01; start and complete transitions in `sprintLifecycle.test.js`	Pass
PMS-F-011	Burndown chart	TC-BURN-01	Pass
PMS-F-012	KPI dashboard	TC-DASH-01	Pass
PMS-F-013	Team workload	TC-WORK-01	Pass
PMS-F-014	Audit trail	TC-LOG-01	Pass
PMS-F-015	Supabase synchronisation	TC-SUPA-01, TC-SET-01	**Blocked** (Section 6)
PMS-F-016	Offline mode	TC-OFF-01, TC-AVAIL-01	Pass
PMS-F-017	Filtering	TC-FILT-01	**Fail** in the main run (DEF-04, low); **Pass** on retest of the fix, PR #98
PMS-F-018	CI/CD pipeline	GitHub Actions history (Unit Test Report, Section 9)	Pass
PMS-NF-001	Performance, under 50 ms	TC-PERF-01 (p95 6.6 ms), TC-LOAD-01 (p95 8.9 ms with 106 tasks)	Pass
PMS-NF-002	Availability through offline fallback	TC-AVAIL-01	Pass

SRS ID	Requirement	Verified by	Result
PMS-NF-003	Usability: contrast, Lighthouse score above 90	No case in the specification; keyboard and screen-reader behaviour covered by `accessibility.test.jsx`	**Not measured** (Section 11)
PMS-NF-004	Reliability: zero data loss across reloads and tab closures	TC-OFF-01, TC-RECOV-01, refresh steps in TC-KAN-02 and TC-TASK-01; TC-SESS-01 (supplementary, simulated Supabase server, Section 12.2)	Pass
PMS-NF-005	Maintainability: zero ESLint warnings, build always passes	`eslint --max-warnings=0`: 0 errors, 0 warnings; `npm run build`: succeeded (143 modules) on the same commit	Pass
PMS-SR-001	Row Level Security	SEC-RLS	**Blocked** (Section 6)
PMS-SR-002	No hardcoded credentials	SEC-KEYS	Pass
PMS-SR-003	HTTPS / TLS 1.3	TC-SEC-01	**Blocked** (Section 6)
PMS-SR-004	XSS input sanitisation	SEC-XSS, SEC-FUZZ	Pass
PMS-SR-005	Dependency audit in CI	Dependency Security Audit workflow: 10 of 10 runs passed; `npm audit --audit-level=high`: 0 vulnerabilities	Pass
(Test Plan only)	In-app documentation viewer	TC-DOCS-01	Pass
(Test Plan only)	Cross-browser support	TC-COMPAT-01	Pass
(Test Plan only)	100+ task records	TC-LOAD-01	Pass
(Test Plan only)	Session token handling	SEC-TOKEN	Pass

Summary against the SRS:

- Functional (PMS-F-001 to 018): in the main run 16 pass, 1 failed with a low-severity defect (now fixed and retested) and 1 is blocked.
- Non-functional (PMS-NF-001 to 005): 4 pass and 1 was not measured.
- Security (PMS-SR-001 to 005): 3 pass and 2 are blocked.

Every SRS requirement has at least one test case or piece of objective evidence.

9. Cross-Browser Compatibility (TC-COMPAT-01)

TC-KAN-01, TC-KAN-02 and TC-TASK-01 were run in each browser in a fresh context, and the position and width of every column were compared with Chrome.

Browser	Version	TC-KAN-01	TC-KAN-02	TC-TASK-01	Layout difference from Chrome
Google Chrome	157.0.8103.38	Pass	Pass	Pass	–
Microsoft Edge	156.0.4314.58	Pass	Pass	Pass	0 px
Firefox (Playwright build)	142.0.1	Pass	Pass	Pass	0 px

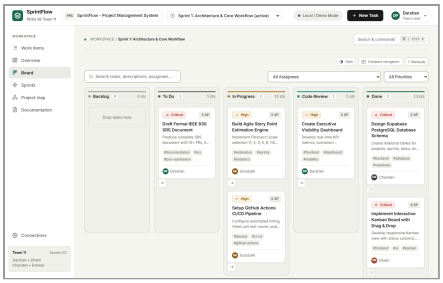


Figure 3: Chrome 157

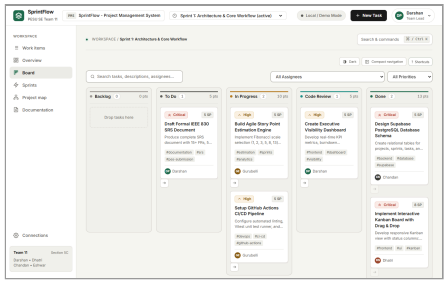


Figure 4: Edge 156

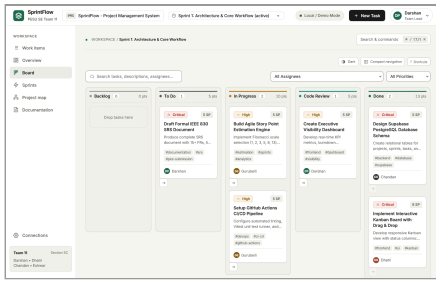


Figure 5: Firefox 142

The board after sign-in in each browser, same build and seed data: five columns and six cards in the same positions. Captured again on 06-11-2026 after the run, because the first Chrome capture was taken while the board was still fading in. Full-size images are in `docs/test-reports/evidence/`.

10. Security Validation (STP Section 5.4)

STP check	Case	Method	Result
XSS input sanitisation	SEC-XSS	`<img onerror>`, `<script>` and `<svg onload>` payloads in a task title, task description and sprint goal; a probe variable shows whether any handler ran	Pass: nothing ran; no injected element in the page
Row Level Security	SEC-RLS	Cross-project REST call from a second account	Blocked (Section 6); policies verified statically by `databaseSecurity.test.js`
TLS 1.3 enforcement	TC-SEC-01	Inspect live traffic	Blocked (Section 6); the client refuses `http://` project URLs
Session token handling	SEC-TOKEN	Scan LocalStorage and SessionStorage after sign-in	Pass: neither the password nor any JWT is stored in zero-configuration mode
API key exposure	SEC-KEYS	Scan the production bundle for complete JWTs and Supabase secret keys	Pass: none in `dist/assets`
Input validation fuzzing	SEC-FUZZ	5,000-character title; Unicode, quotes, `&`, SQL-like text and emoji	Pass: title bounded to 120 characters with no crash; special characters stored and shown as typed

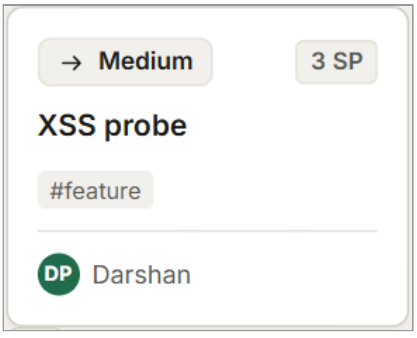


Figure 6: SEC-XSS: the payload in the title is shown as plain text and the image handler never ran.

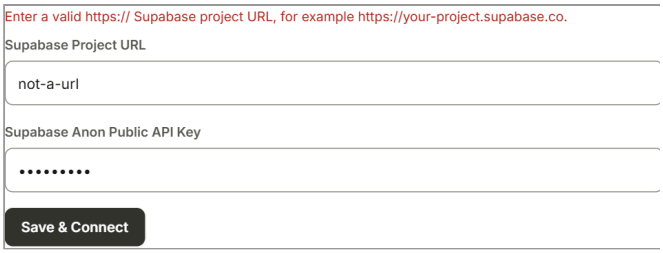


Figure 7: TC-SET-01 step 3 / TC-SEC-01: a malformed or http:// URL is refused.

11. Exit Criteria Assessment (STP Section 5.3)

Exit criterion	Status	Detail
100% of planned test cases executed	**Not met**	23 of the 26 specified cases ran in full. TC-SUPA-01, TC-SET-01 (steps 1–2) and TC-SEC-01 wait on a Supabase test project (#96).
0 critical or high-severity defects open	Met	Two defects were found, neither critical nor high, and both are fixed and retested: DEF-04 (PR #98) and DEF-05 (PR #101).

Exit criterion	Status	Detail
All functional requirements verified against acceptance criteria	Partly met	17 of 18 SRS functional requirements pass once the DEF-04 fix is counted; PMS-F-015 (Supabase sync) is blocked.
All non-functional requirements measured and within threshold	Partly met	4 of 5 SRS non-functional requirements were measured, all within threshold. PMS-NF-003 (Lighthouse score above 90) has no test case and was not measured.
RTM shows complete coverage	Met	Every SRS requirement maps to a test case or objective evidence (Section 8).
Test Summary Report reviewed and approved	Met	Reviewed and merged into `main` together with the other reports (PR #97, 06-11-2026).

To close the remaining criteria: run the Section 6 cases on a Supabase test project (#96), and add a Lighthouse accessibility measurement for PMS-NF-003 to the Test Case Specification.

12. Regression on the Final Build

After the main run, four pull requests were merged into `main` on 06-11-2026: #98 (the DEF-04 fix), #99 (Dhatri's second-machine pre-check), #101 (the DEF-05 fix, a tooltip warning and the CI Node.js version) and #85 (verified cloud sessions, merged after review fixes). Because #85 changes how sign-in works, we ran the whole suite again on the final `main`.

Run	Build	Browsers	Pass	Fail	Blocked
Main run (Sections 3 to 11)	`f9e0727`	Chrome; Edge and Firefox for TC-COMPAT-01	28	1	4
DEF-04 retest (Section 5.2)	`e8f5905` (PR #98)	Chrome	29	0	4
Final regression, 06-11-2026, 23:05 IST	`2b8fdbc` (final `main`)	Chrome; Edge and Firefox for TC-COMPAT-01	29	0	4

The final results match the main run case for case, except that TC-FILT-01 now passes. TC-COMPAT-01 again found a 0 px layout difference between the three browsers, and the same four cloud cases stay blocked. On the same commit all 259 unit tests pass (Unit Test Report, Section 10).

12.1 DEF-05 retest

PR #101 makes the panel say "task" for a count of one and adds a unit test for it. On the final build Dhatri and Chandan now read "1 task • 8 story points assigned" and "1 task • 5 story points assigned", while members with two tasks still read "2 tasks". TC-WORK-01 passes, and issue #100 was closed by the merge.

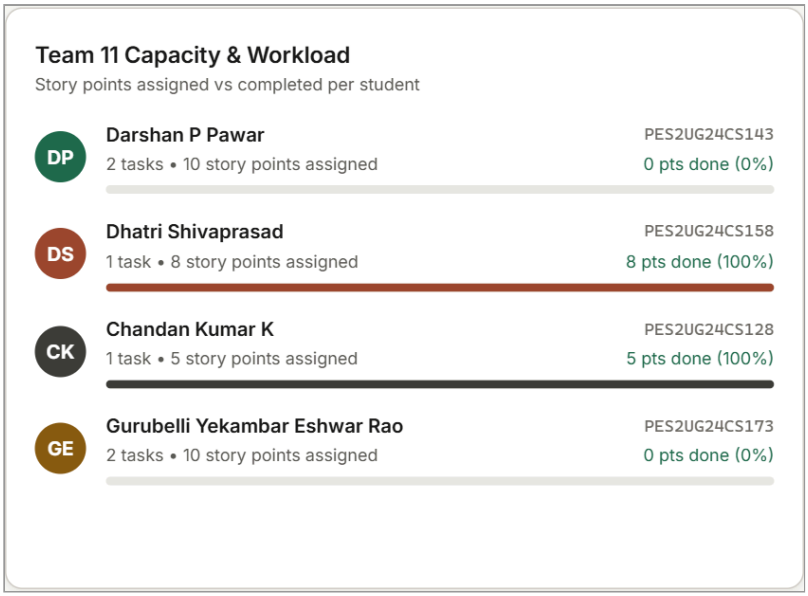


Figure 8: DEF-05 retest on the final build: "1 task" for a member with one task.

12.2 Supplementary case TC-SESS-01: a network failure must not sign out a member

PR #85 re-checks the cloud session whenever Supabase reports an auth event, for example when the browser tab regains focus. While reviewing it we found that a failed check signed the member out even when only the network had failed, which threw away any task being edited. That is a data-loss risk under PMS-NF-004. The fix keeps the member on network errors and signs out only when Supabase rejects the session. No Supabase test project is available yet (Section 6), so we ran this case in Chrome against a simulated Supabase server: Playwright answered the app's Supabase requests, and no real database was used.

Step	Expected result	Before the fix	Final build (`2b8fdbc`)
1. Sign in to the cloud project	The workspace opens	Pass	Pass
2. Type a task title; the session check then fails because the network is down	Still signed in; the dialog and the typed title remain	**Fail**: sent to the sign-in page, draft lost	Pass
3. The network comes back	The session is checked again; still signed in	Not reached	Pass
4. Supabase rejects the session (401)	Signed out with "Your cloud session has ended"	Not reached	Pass

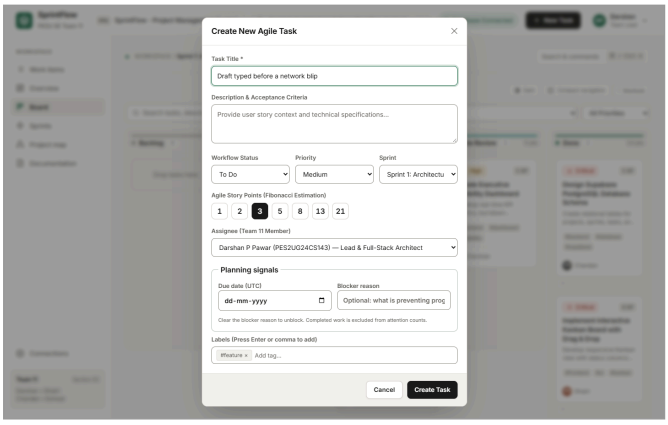


Figure 9: Step 2 on the final build: still signed in, the draft is kept.

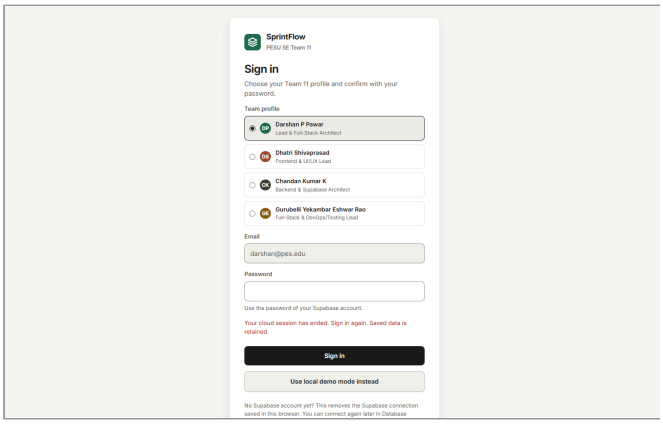


Figure 10: Step 4: a rejected session signs the member out with a clear message.

The same logic is covered by 15 unit tests in `src/tests/sessionVerification.test.js`, which fail on the code before the fix and pass after it. This case does not replace TC-SUPA-01 on a real Supabase project, which is still open

under issue #96.

13. Conclusion

In zero-configuration mode SprintFlow behaves as specified in Chrome, Edge and Firefox. Every specified workflow passed: creating, editing and deleting tasks, drag-and-drop, one-click advance, estimation, sprint planning, the burndown chart, the dashboard, the audit trail, offline use and recovery. Drag-and-drop rendered within 9 ms at the 95th percentile even with 106 tasks on the board, against a 50 ms budget. No XSS payload ran and no credential was exposed in the bundle. Testing found two defects, a low-severity usability defect (DEF-04) and a cosmetic wording defect (DEF-05). Both were fixed, retested and merged on 06-11-2026, and a full regression run on the final build in all three browsers passed with no failures. The only open work is the four cloud-database cases, which need a Supabase test project (#96).

Appendix A: Detailed Results

Each case from the main run with every expected result that was checked and the actual result the browser showed.

TC-AUTH-01: Authenticate and establish session (Pass)

SRS: PMS-F-001 · Test Plan: PMS-F-001 · Browser: Chrome

Expected result checked	Actual result	Verdict
Navigation bar shows the profile name	Darshan	Pass
Navigation bar shows the role	Team Lead	Pass
Session persists after refresh (no sign-in view)	signed in after reload	Pass

TC-PROJ-01: Show project metadata (supplementary) (Pass)

SRS: PMS-F-002 · Test Plan: – · Browser: Chrome

Expected result checked	Actual result	Verdict
Top bar shows the project key and name	SprintFlow PESU SE Team 11 PMS SprintFlow - Project Management System Sprint 1: Architecture & Core Workflow (active) Sp…	Pass
Top bar shows the active sprint	Sprint 1: Architecture & Core Workflow (active)	Pass
Dashboard shows the project key, name and date range	2026-09-01 → 2026-10-30	Pass

TC-TASK-01: Create a task with complete metadata (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card shows the title	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows a High priority badge	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows a 5-point estimate	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
Card shows the assignee	High 5 SP Implement burndown chart #feature DS Dhatri	Pass
With Backlog chosen, the card appears at the top of the Backlog column	Implement burndown chart (Backlog)	Pass
Cards survive a page refresh	both present after reload	Pass

Notes: With the default Workflow Status the card lands in "To Do". SPRINT1_DEVIATIONS.md records that a new task starts in To Do unless another status is chosen; the specification text still says Backlog.

TC-TASK-02: Assign Critical priority (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-016 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card shows a Critical badge	Critical	Pass
Badge carries a text label, not colour alone	Critical	Pass

TC-TASK-03: Reject a task with an empty title (Pass)

SRS: PMS-F-006 · Test Plan: PMS-F-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Empty title is rejected with a validation message	browser validation: "Please fill out this field."	Pass
Single-space title is rejected with a validation message	inline message: "Enter a task title and choose a valid priority."	Pass
No task is created in either case	6 cards before and after	Pass

TC-LABEL-01: Add labels to a task (supplementary) (Pass)

SRS: PMS-F-007 · Test Plan: – · Browser: Chrome

Expected result checked	Actual result	Verdict
Labels show as separate pills, lower-cased and de-duplicated	#feature #ui #api	Pass
Labels survive a page refresh	#feature #ui #api	Pass

TC-KAN-01: Render the five-column board (Pass)

SRS: PMS-F-003 · Test Plan: PMS-F-003 · Browser: Chrome

Expected result checked	Actual result	Verdict
Exactly five columns in lifecycle order	Backlog, To Do, In Progress, Code Review, Done	Pass
Backlog shows its task count	count badge 0, cards 0	Pass
To Do shows its task count	count badge 1, cards 1	Pass
In Progress shows its task count	count badge 2, cards 2	Pass
Code Review shows its task count	count badge 1, cards 1	Pass
Done shows its task count	count badge 2, cards 2	Pass

TC-KAN-02: Transition a task by drag-and-drop (Pass)

SRS: PMS-F-004 · Test Plan: PMS-F-004 · Browser: Chrome

Expected result checked	Actual result	Verdict
Card renders in In Progress on release	In Progress	Pass
Source column count decreases by one	1 → 0	Pass
Target column count increases by one	2 → 3	Pass
New state survives a refresh	In Progress	Pass

TC-KAN-03: Advance a task by single action (Pass)

SRS: PMS-F-005 · Test Plan: PMS-F-005 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: task is in To Do	To Do	Pass
Task moves to In Progress	In Progress	Pass
A task in Done cannot advance further	0 enabled advance controls in Done	Pass

TC-KAN-04: Cancel a drag outside any column (Pass)

SRS: PMS-F-004 · Test Plan: PMS-F-004 · Browser: Chrome

Expected result checked	Actual result	Verdict
A drag was actually started	1 dragstart, 0 drop events	Pass
Card returns to its original column	In Progress → In Progress	Pass
Status is unchanged after refresh	In Progress	Pass

TC-FILT-01: Filter by priority (Fail)

SRS: PMS-F-017 · Test Plan: PMS-F-011 · Browser: Chrome

Expected result checked	Actual result	Verdict
Only High priority tasks are visible	3 visible: high, high, high	Pass
No task matches the combined filters	0 cards visible	Pass
An explicit empty state names the active filters	five columns each show "Drop tasks here"; no message names the active filters	Fail
The empty state offers to clear the filters	"Reset Filters" button shown	Pass

TC-EST-01: Assign Fibonacci story points (Pass)

SRS: PMS-F-008 · Test Plan: PMS-F-006 · Browser: Chrome

Expected result checked	Actual result	Verdict
Only the seven Fibonacci values are offered	1, 2, 3, 5, 8, 13, 21	Pass
No free numeric estimate field exists	0 numeric inputs	Pass
Estimate 1 persists on the card	1 SP	Pass
Estimate 2 persists on the card	2 SP	Pass
Estimate 3 persists on the card	3 SP	Pass
Estimate 5 persists on the card	5 SP	Pass
Estimate 8 persists on the card	8 SP	Pass
Estimate 13 persists on the card	13 SP	Pass
Estimate 21 persists on the card	21 SP	Pass
The final estimate is reflected in sprint capacity	33 → 49 (task 5 → 21)	Pass

TC-EST-02: Calculate sprint capacity (Pass)

SRS: PMS-F-009 · Test Plan: PMS-F-008 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: the new sprint has no tasks	0	Pass
Capacity reports 16 points for tasks of 3, 5 and 8	16	Pass
Capacity is recomputed to 11 after removing the 5-point task	11	Pass

TC-SPR-01: Create a sprint (Pass)

SRS: PMS-F-010 · Test Plan: PMS-F-007 · Browser: Chrome

Expected result checked	Actual result	Verdict
Sprint appears in the sprint list	1	Pass
Sprint shows its goal	PLANNED 2026-11-06 → 2026-11-19 Sprint 3 View sprint Start Sprint Delete Sprint Sprint Goal: Deliver estimation engine 0 Planned Points 0 Completed 0 Remaining …	Pass
Sprint shows its two-week date range	2026-11-06 → 2026-11-19	Pass
Sprint is available as a task assignment target	Backlog (No Sprint) | Sprint 1: Architecture & Core Workflow | Sprint 2: Analytics & Release Candidate | Sprint 3	Pass

TC-BURN-01: Update the burndown chart (Pass)

SRS: PMS-F-011 · Test Plan: PMS-F-009 · Browser: Chrome

Expected result checked	Actual result	Verdict
Precondition: today has a recorded actual value	20	Pass
Task moved to Done	Done	Pass
Actual line falls by exactly the completed points	20 → 15 (task 5 SP)	Pass
Ideal line is unchanged	19 ideal values compared	Pass

Notes: Read from the chart's "Daily burndown values" table, which lists the same series the SVG draws.

TC-DASH-01: Display KPI metrics (Pass)

SRS: PMS-F-012 · Test Plan: PMS-F-010 · Browser: Chrome

Expected result checked	Actual result	Verdict
Progress, velocity and workload are shown	total 33, delivered 13, completion 39%, remaining 20	Pass
Total matches the board	board 33 pts, dashboard 33	Pass
Delivered velocity matches the Done column	Done 13 pts, dashboard 13	Pass
Progress equals completed points / total points	13/33 = 39.4%, shown 39%	Pass

TC-WORK-01: Show workload distribution (Pass)

SRS: PMS-F-013 · Test Plan: PMS-F-017 · Browser: Chrome

Expected result checked	Actual result	Verdict
Tasks are assigned across at least three members	10, 8, 5, 10	Pass
Member totals sum to the sprint total	10 + 8 + 5 + 10 = 33; sprint total 33	Pass

TC-DOCS-01: Render in-app documentation (Pass)

SRS: not in SRS · Test Plan: PMS-F-018 · Browser: Chrome

Expected result checked	Actual result	Verdict
"Functional Requirements" tab renders readable content	3172 characters	Pass
"Non-Functional" tab renders readable content	1093 characters	Pass
"Traceability Matrix" tab renders readable content	947 characters	Pass
Documentation has a readable heading	Software Requirements Specification (SRS) & RTM	Pass

TC-SUPA-01: Persist to Supabase PostgreSQL (Blocked)

SRS: PMS-F-015 · Test Plan: PMS-F-013 · Browser: Chrome

Blocked: Precondition not met: needs a Supabase test project with database/schema.sql (and the 20261103 migration) applied.

TC-OFF-01: Persist offline without configuration (Pass)

SRS: PMS-F-016 · Test Plan: PMS-F-014 · Browser: Chrome

Expected result checked	Actual result	Verdict
Both tasks are restored after reopening the tab	2 of 2 present	Pass
Interface indicates offline (LocalStorage) persistence	Local / Demo Mode	Pass

TC-SET-01: Switch data mode from settings (Blocked)

SRS: PMS-F-015 · Test Plan: PMS-F-015 · Browser: Chrome

Blocked: Steps 1-2 need valid credentials for a provisioned Supabase test project; only step 3 (negative case) was executed.

Expected result checked	Actual result	Verdict
Step 3: a malformed URL is rejected with a clear message	Enter a valid https:// Supabase project URL, for example https://your-project.	Pass
Step 3: the previous (offline) configuration is retained	Local / Demo Mode	Pass

TC-LOG-01: Record an audit trail for mutations (Pass)

SRS: PMS-F-014 · Test Plan: PMS-F-012 · Browser: Chrome

Expected result checked	Actual result	Verdict
Exactly three new audit entries	4 → 7	Pass
Entries are CREATE, UPDATE, DELETE in chronological order	CREATE, UPDATE, DELETE	Pass
Each entry carries a timestamp	2026-11-06T12:08:22.068Z, 2026-11-06T12:08:22.563Z, 2026-11-06T12:08:23.106Z	Pass
Timestamps are in chronological order	non-decreasing	Pass
Each entry names the acting user and the task	CREATE Darshan P Pawar created task TC-LOG-01 audit probe → To Do || UPDATE Darshan P Pawar updated task TC-LOG-01 audit probe (renamed) || DELETE Darshan P Pawar deleted task TC-LOG-01 audit probe (renamed)…	Pass

TC-PERF-01: Drag-and-drop transition under 50 ms (Pass)

SRS: PMS-NF-001 · Test Plan: PMS-NF-001 · Browser: Chrome

Expected result checked	Actual result	Verdict
Drop-to-render p95 under 50 ms	6.6 ms	Pass

Notes: 10 measured drops after 2 warm-ups: p50 2.3 ms, p95 6.6 ms, max 6.6 ms.

TC-AVAIL-01: Stay usable with the network disabled (Pass)

SRS: PMS-NF-002 · Test Plan: PMS-NF-002 · Browser: Chrome

Expected result checked	Actual result	Verdict
Application stays usable with the network disabled	task created and all views opened	Pass
Application reports offline (local) mode	Local / Demo Mode	Pass

TC-COMPAT-01: Core workflows on Chrome, Edge and Firefox (Pass)

SRS: not in SRS · Test Plan: PMS-NF-003 · Browser: Chrome, Edge and Firefox

Expected result checked	Actual result	Verdict
chrome: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
edge: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
firefox: TC-KAN-01, TC-KAN-02 and TC-TASK-01 pass	TC-KAN-01 pass, TC-KAN-02 pass, TC-TASK-01 pass	Pass
edge: column layout matches chrome within 4 px	max drift 0 px	Pass
firefox: column layout matches chrome within 4 px	max drift 0 px	Pass

Notes: Browser versions: chrome 157.0.8103.38, edge 156.0.4314.58, firefox 142.0.1.

TC-SEC-01: Encrypted transport to Supabase (Blocked)

SRS: PMS-SR-003 · Test Plan: PMS-NF-004 · Browser: Chrome

Blocked: Inspecting live HTTPS traffic needs a connected Supabase project. Client-side enforcement was verified instead.

Expected result checked	Actual result	Verdict
A plain http:// project URL is refused	Enter a valid https:// Supabase project URL, for example https://your-project.	Pass

TC-LOAD-01: Board with 100+ tasks (Pass)

SRS: not in SRS · Test Plan: PMS-NF-005 · Browser: Chrome

Expected result checked	Actual result	Verdict
All seeded tasks render on the board	106 cards	Pass
Every seeded task is accessible as a control	100 task controls	Pass
Drag-and-drop stays responsive (p95 under 50 ms with 106 tasks)	8.9 ms	Pass

Notes: Board ready 162 ms after navigation with 106 tasks; drop-to-render p95 8.9 ms over 5 drops.

TC-RECOV-01: No data loss across disconnect and reconnect (Pass)

SRS: PMS-NF-004 · Test Plan: PMS-NF-006 · Browser: Chrome

Expected result checked	Actual result	Verdict
No task is lost across disconnect and reconnect	2 of 2 present	Pass

Notes: Zero-config mode. The cloud retry queue across a reconnect is covered by unit tests (persistenceRecovery, recovery).

SEC-XSS: XSS input sanitization (Pass)

SRS: PMS-SR-004 · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
No injected script or handler executes	window.__sfXss never set	Pass
Input is rendered as text, not markup	0 injected elements in cards	Pass

SEC-RLS: Row Level Security on direct API calls (Blocked)

SRS: PMS-SR-001 · Test Plan: STP 5.4 · Browser: Chrome

Blocked: Needs a live Supabase project to send cross-project API calls. The policies in database/schema.sql are checked statically by src/tests/databaseSecurity.test.js.

SEC-TOKEN: Session token handling (Pass)

SRS: not in SRS · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
The password is not kept in browser storage	not found	Pass
No access token is kept in zero-config mode	no JWT in storage	Pass

Notes: Cloud JWT expiry and rejection need a live Supabase project (see TC-SEC-01).

SEC-KEYS: API key exposure in the client bundle (Pass)

SRS: PMS-SR-002 · Test Plan: STP 5.4 · Browser: none (static scan of the build)

Expected result checked	Actual result	Verdict
No complete JWT (anon or service_role) is embedded in the bundle	0 found in 2 files	Pass
No Supabase secret key is embedded in the bundle	0 found	Pass

Notes: Static scan of the production bundle in dist/assets.

SEC-FUZZ: Input validation fuzzing (Pass)

SRS: PMS-SR-004 · Test Plan: STP 5.4 · Browser: Chrome

Expected result checked	Actual result	Verdict
An over-long title is bounded, not crashing the app	120 characters stored	Pass
Special characters are stored and shown as typed	Ω ✓ "quotes" & ampersand; DROP TABLE tasks; -- 🚀	Pass

Appendix B: Screenshot Evidence

These screenshots were captured on 06-11-2026 after the main run by repeating each case's steps in a fresh Chrome session on the same build (`f9e0727`).

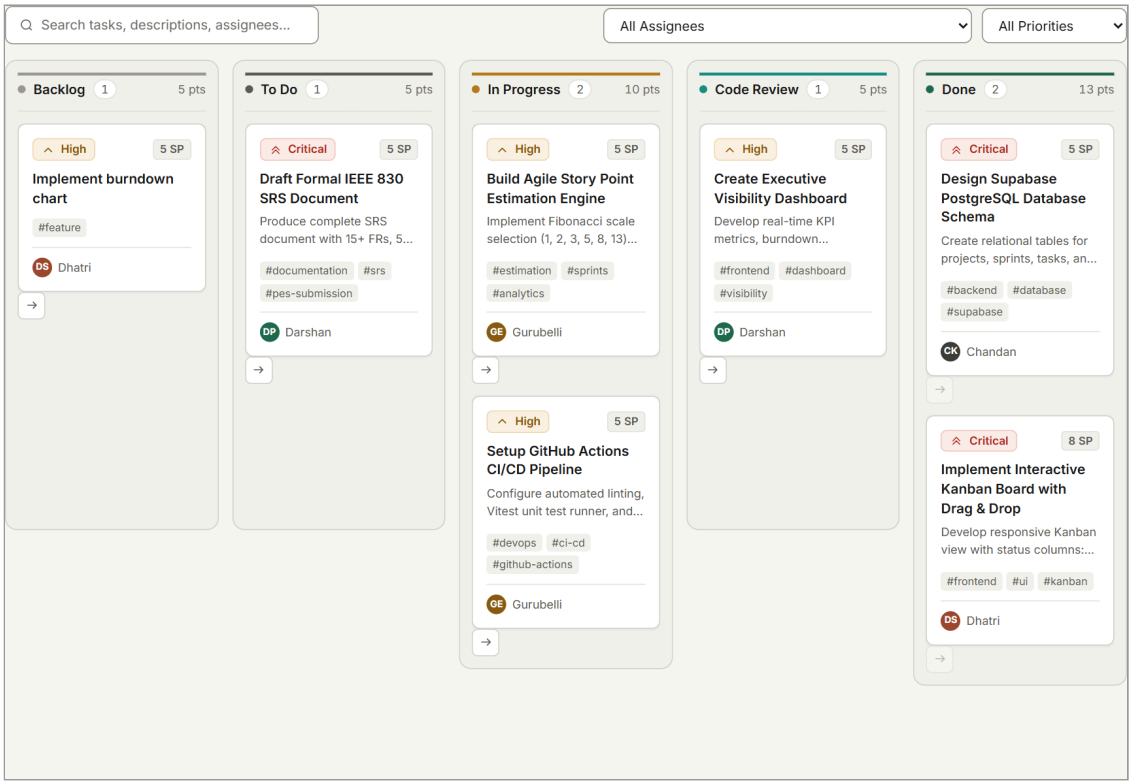


Figure 11: TC-TASK-01: "Implement burndown chart" (High, 5 SP, Dhatri) at the top of Backlog after a page refresh.

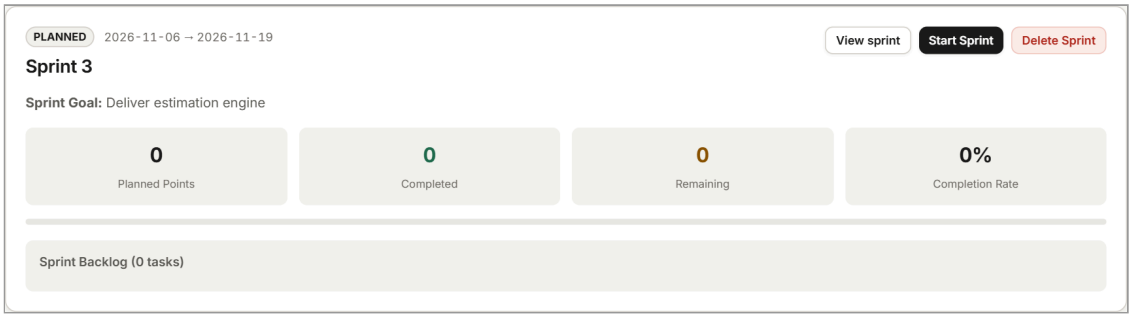


Figure 12: TC-SPR-01: Sprint 3 with its goal and two-week date range, in the Planned state.

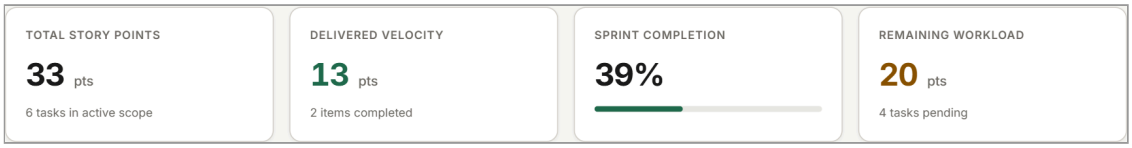


Figure 13: TC-DASH-01: KPI cards (total 33 points, 13 delivered, 39% complete, 20 remaining).

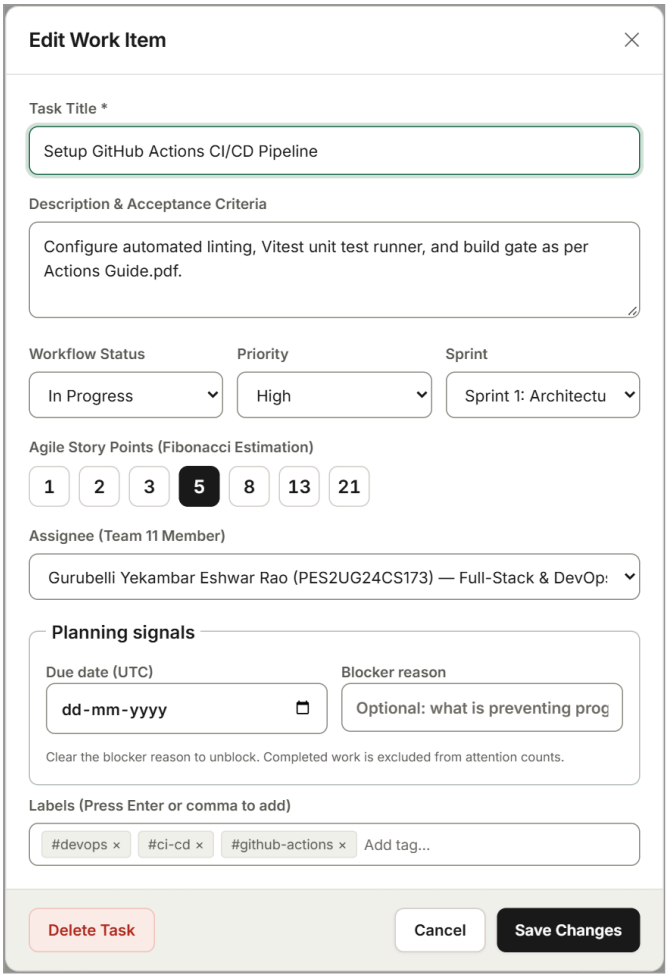


Figure 14: TC-EST-01: the task dialog offers only 1, 2, 3, 5, 8, 13 and 21.

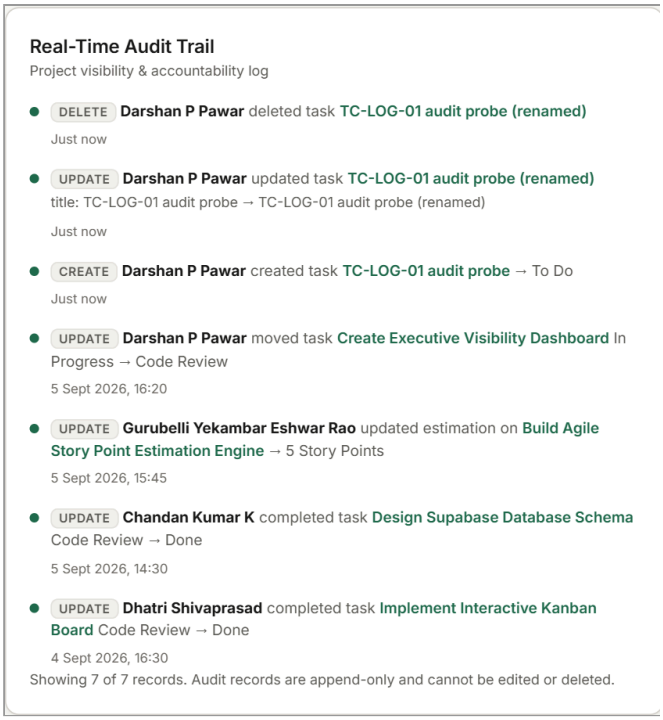


Figure 15: TC-LOG-01: CREATE, UPDATE and DELETE records with the user, task and time.

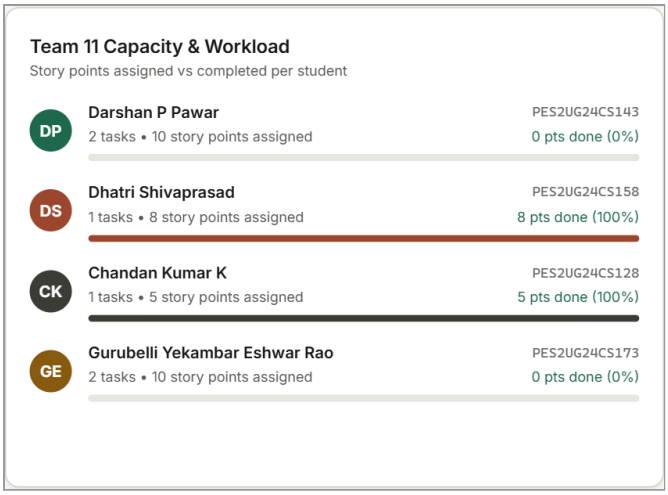


Figure 16: TC-WORK-01: points per member add up to the sprint total (note "1 tasks", DEF-05).

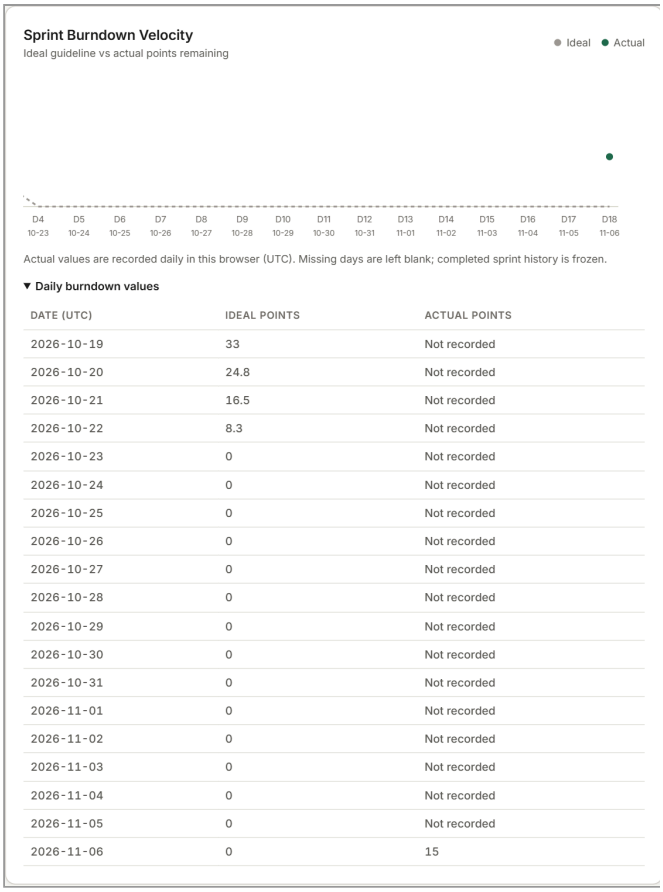


Figure 17: TC-BURN-01: today's actual value (15) after a 5-point task was completed, shown on the chart and in its table.

Appendix C: Re-running the Suite

```
npm ci
npx playwright install firefox     # once; Chrome and Edge use the installed browsers
npm run build
npm run preview                    # terminal 1: serves http://localhost:4173/
npm run test:system                # terminal 2: writes test-results/system/
```

The build must be made without Supabase credentials (no `.env` file), so that no test writes to a real database. `BROWSERS=chrome` runs Chrome only, and `ONLY=TC-KAN-02,TC-PERF-01` re-runs chosen cases.

Two behaviours of browser automation shaped the script; neither is a SprintFlow defect. Under Chrome automation a card that has already been dragged once does not start a second drag, so each timing sample drags a different card. Automation also cannot scroll during a drag, so drops land on the target column at the dragged card's height. Firefox has neither limitation.